Medical Question Answering on PubMedQA: Analysing Text Representations and Classifier Design

Group 16
EMATM0067 – Introduction to AI and Text Analytics
University of Bristol

Abstract

This report investigates how different text representations and AI classification methods affect the quality of answers to medical questions on the PubMedQA dataset. We design and evaluate a complete question-answering classification pipeline, exploring two primary design axes: (1) text representation, comparing TF-IDF n-grams, BioBERT sentence embeddings, and varying input compositions; and (2) classifier choice, comparing Logistic Regression, calibrated Linear SVM, Multinomial Naive Bayes, SMOTE-augmented models, a soft voting ensemble, zero-shot LLM classification, and fine-tuned BioBERT. Our best model, a fine-tuned BioBERT classifier with class-weighted loss, achieves a test macro-F1 of 0.4108. Extensive error analysis reveals that the *maybe* class is misclassified 81.8% of the time, and 24% of test samples are misclassified by all models, representing genuinely ambiguous cases. A quality predictor analysis confirms that surface-level text features cannot reliably predict answer correctness.

1 Introduction

[1]

Biomedical question answering (QA) systems aim to help clinicians and researchers find answers to clinical questions without manually searching through thousands of publications (1). As the volume of biomedical literature continues to grow rapidly, automated QA becomes increasingly important for evidence-based medicine and clinical decision support (2). The BioASQ challenge (3) and subsequent benchmarks have driven substantial progress in this area, yet robust three-class answer classification remains an open challenge.

The PubMedQA dataset (4) provides a benchmark for this task, consisting of 1,000 expert-annotated biomedical questions drawn from PubMed abstracts. Each instance contains a research question, one or more context paragraphs from the corresponding abstract, a long-form answer, and a final decision label: *yes*, *no*, or *maybe*. Unlike binary sentiment classification, the *maybe* class captures genuine scientific uncertainty where evidence is insufficient or inconclusive.

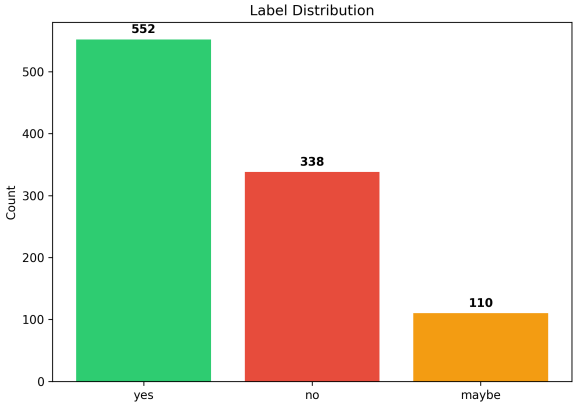

Figure 1: PubMedQA label distribution across 1,000 expert-annotated instances. The dataset is imbalanced: *yes* dominates at 55.2%, while *maybe* accounts for only 11%.

A key challenge in this dataset is the severe class imbalance. As shown in Figure 1, the distribution is heavily skewed: 552 instances are labelled *yes* (55.2%), 338 are *no* (33.8%), and only 110 are *maybe* (11.0%). This imbalance directly impacts model training, as classifiers tend to over-predict the majority class.

Furthermore, Figure 2 shows that the majority of instances contain 3–4 context sentences, with a long tail extending to 10 sentences, indicating variable evidence density across the dataset. Questions average approximately 10–15 words, while contexts average around 200 words with a maximum of 398. This asymmetry between short questions and long contexts poses representation challenges for traditional bag-of-words methods (5).

[1] GitHub link: http:\

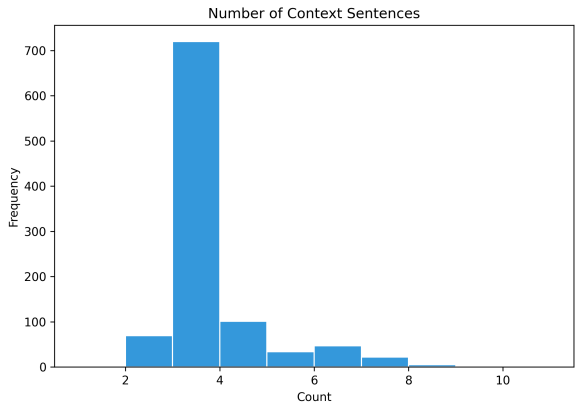

Figure 2: Distribution of context sentence counts. Most instances contain 3–4 context sentences, with a long tail up to 10 sentences per instance.

Recent advances in domain-specific pre-training (2; 6) and transfer learning (7) have demonstrated the value of leveraging biomedical corpora for NLP tasks. BioBERT (2), pre-trained on PubMed abstracts and PMC full-text articles, has achieved strong results across biomedical benchmarks, while retrieval-augmented (8) and text-to-text approaches (9) offer promising alternatives. Sequence tagging methods (10) provide an alternative extractive paradigm for biomedical QA.

This report systematically investigates two design axes: (1) how text representation affects classification quality, and (2) how classifier choice handles class imbalance and decision complexity. We additionally explore classification versus zero-shot LLM-based generation, implement a quality predictor as requested in the task brief, and conduct detailed error analysis.

1.1 Related Work

Biomedical QA has evolved substantially in recent years. Early approaches relied on information retrieval combined with rule-based answer extraction (3). The introduction of transformer-based models marked a significant improvement, with BioBERT (2) achieving state-of-the-art results by pre-training on PubMed abstracts. ClinicalBERT (5) further demonstrated the value of domain-specific pre-training by focusing on clinical notes, while PubMedBERT (6) showed that training from scratch on biomedical text outperforms continued pre-training from general-domain checkpoints.

For PubMedQA specifically, Jin et al. (4) established baseline results showing that even strong language models struggle with the three-class formulation, particularly the *maybe* class. Transfer learning studies (7) have systematically evaluated BERT and ELMo across biomedical benchmarks, finding that domain-specific pre-training consistently outperforms general-domain models. Text-to-text models such as SciFive (9) frame QA as a generation task, while BIOREADER (8) incorporates external evidence through retrieval augmentation.

2 Methods

2.1 Baseline System

Our baseline pipeline consists of a TF-IDF vectoriser with unigram and bigram features (max 30,000 features) fed into a Logistic Regression (LR) classifier with balanced class weights. The input text is the concatenation of the cleaned question and context, delimited by a `[SEP]` token. Text preprocessing involves lowercasing, removal of irregular punctuation via regex, and whitespace normalisation. Labels are encoded as integers: no = 0, maybe = 1, yes = 2.

The data is split into training (720 samples, 72%), development (80 samples, 8%), and test (200 samples, 20%) sets using stratified sampling to preserve label proportions. All reported metrics use 5-fold stratified cross-validation (CV) on the training set, with final evaluation on the held-out test set.

We additionally engineer handcrafted NLP features to capture linguistic patterns: negation cues (*no*, *not*, *without*, *lack*), causal language (*cause*, *effect*, *impact*, *lead to*), comparison terms (*compared*, *versus*, *better*), association language (*associated*, *correlated*), risk vocabulary, hedging indicators (*may*, *suggest*, *possible*), and context evidence strength scores. These features supplement the main pipeline and are later used by the quality predictor.

2.2 Axis 1: Text Representation

We hypothesise that the information fed to the classifier strongly affects performance. Four strategies are compared:

A1a (Question Only): Using only the question text. We expect this to perform worst, as the answer depends on evidence in the context.

A1b (Question + Context): The baseline input. Adding context should help by providing evidence

phrases associated with each label.

A1c (Question + Context + Long Answer): The richest input. The long answer often contains decisive hedging or affirmative language.

A1d (BioBERT Sentence Embeddings): Dense 768-dimensional embeddings from a pre-trained BioBERT sentence transformer (2) on Q+Ctx input, fed into LR.

2.3 Axis 2: Classifier Comparison

Holding the text representation fixed (combined word-level and character-level TF-IDF on Q+Ctx, yielding ∼70,000 features), we compare multiple classifiers with hyperparameter tuning via GridSearchCV:

A2a (Tuned Logistic Regression): Grid search over $C \in \{0.05, 0.1, 0.5, 1.0, 2.0, 5.0\}$ with balanced class weights and the liblinear solver.

A2b (Calibrated Linear SVM): Linear SVM tuned over C values, calibrated using Platt scaling, with a custom *maybe*-class probability threshold optimised on the development set.

A2c (Multinomial Naive Bayes): Tuned over smoothing parameter $\alpha \in \{0.05, 0.1, 0.25, 0.5, 1.0\}$.

A2d (SMOTE + LR): SMOTE applied to the training set only, equalising all three classes to 398 samples each, followed by tuned LR.

A2e (Soft Voting Ensemble): Combining calibrated SVM, tuned LR, and MNB via soft voting (probability averaging).

2.4 Classification vs. LLM-Based Generation

Following the task specification, we compare supervised classification against a zero-shot approach using `facebook/bart-large-mnli`, a model trained on natural language inference (NLI). The pipeline classifies each Q+Ctx test sample into candidate labels {*yes*, *no*, *maybe*} without any task-specific training.

2.5 Fine-tuned BioBERT

We fine-tune BioBERT (2) (`dmis-lab/biobert-base-cased-v1.2`) for 3-class sequence classification on Q+Ctx inputs. Training uses class-weighted cross-entropy loss with inverse-frequency weights to penalise minority-class misclassification. We use the AdamW optimiser with learning rate 2×10^{-5}, 10% linear warmup, batch size 16, max sequence length 256, 5 epochs, and early stopping with patience 2. The best checkpoint is selected by validation macro-F1.

2.6 Quality Predictor

As requested in the task brief, we investigate whether a model can predict if a question is likely to produce a good-quality answer. We use 11 handcrafted features (question length, context length, long answer length, number of contexts, negation, causal language, comparison, association, risk, evidence strength, hedging) and train three models: Random Forest (400 trees), MLP (64→32 hidden units, ReLU), and Logistic Regression, all evaluated via 5-fold stratified CV.

3 Experimental Evaluation

3.1 Performance Metrics and Setup

We report **macro-averaged F1 score** as our primary metric, as it weights all three classes equally regardless of frequency. This is essential given the class imbalance: a model predicting *yes* for all samples achieves 55% accuracy but only 0.24 macro-F1. Accuracy is reported as a secondary metric. Core software: Scikit-learn, Hugging Face Transformers, imbalanced-learn, and SentenceTransformers.

3.2 Baseline Results

The baseline TF-IDF (1,2)-grams + LR achieved a test accuracy of 47% and test macro-F1 of 0.3306. Figure 3 shows the confusion matrix: the baseline heavily over-predicts *yes*, correctly classifying 78 of 110 *yes* samples but misclassifying 51 of 68 *no* samples as *yes* and 14 of 22 *maybe* samples as *yes*. Only 2 *maybe* samples are correctly predicted, while 6 are misclassified as *no* and 14 as *yes*. The 5-fold CV macro-F1 on the training set was 0.37, indicating a modest generalisation gap. This over-prediction of the majority class confirms that class imbalance is a dominant factor, and the textual cues distinguishing *maybe* from *yes* are subtle hedging phrases embedded within otherwise affirmative contexts.

3.3 Axis 1 Results: Text Representation

Figure 4 presents the 5-fold CV macro-F1 with standard deviation bars for the three TF-IDF text variants. All three achieve similar scores around

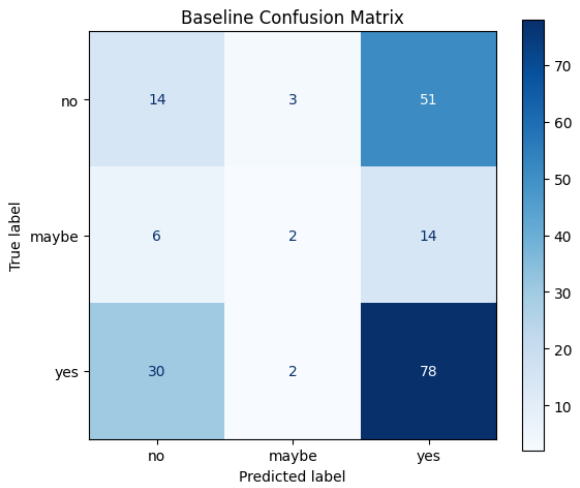

Figure 3: Baseline confusion matrix (TF-IDF + LR on Q+Ctx). The model strongly over-predicts the majority *yes* class, with only 2 correct *maybe* predictions out of 22.

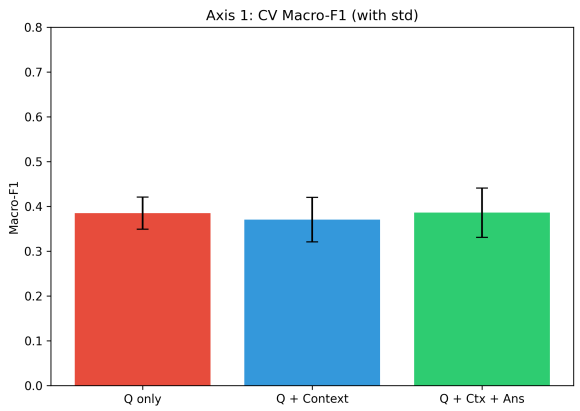

Figure 4: Axis 1: 5-fold CV macro-F1 with standard deviation. All three text variants achieve overlapping scores (∼0.37–0.39), indicating no significant difference.

0.37–0.39 with overlapping error bars, indicating no statistically significant difference between input compositions under a TF-IDF model.

Figure 5 confirms this on the test set: Q only achieves 0.3302, Q+Context (baseline) 0.3306, and Q+Ctx+Ans 0.3243. Contrary to our hypothesis, adding context or the long answer does not improve performance for TF-IDF models. The additional text introduces as much noise as signal—context paragraphs contain 150–250 words of dense biomedical jargon that dilutes discriminative n-grams. BioBERT embeddings (A1d) achieved a test macro-F1 of 0.3637, the best Axis 1 result, offering a modest improvement through domain-specific semantic representations but suggesting the bottleneck lies in the classification task itself.

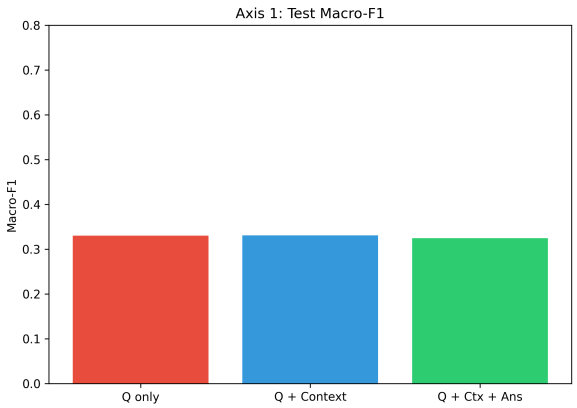

Figure 5: Axis 1: Test macro-F1. All three TF-IDF variants perform comparably (∼0.33), confirming that adding more text does not help bag-of-words models.

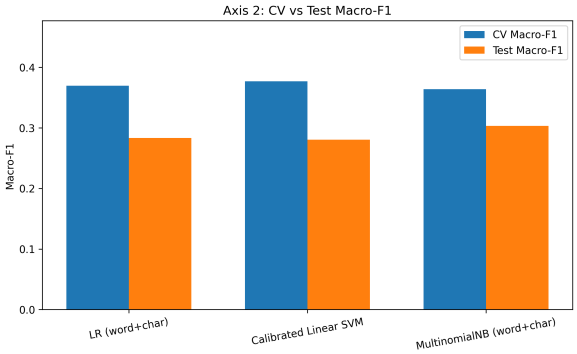

Figure 6: Axis 2: CV vs. test macro-F1. A substantial generalisation gap exists between CV (∼0.37) and test (∼0.28–0.30) scores for all classifiers.

3.4 Axis 2 Results: Classifier Comparison

Figure 6 shows the traditional classifier comparison. All three classifiers achieve similar CV macro-F1 (∼0.37), but test scores drop: LR 0.2831, calibrated SVM 0.2806, MNB 0.3033. This CV-to-test gap (∼0.08–0.10) indicates overfitting to the training distribution.

Figure 7 shows the MNB confusion matrix—the best Axis 2 classifier by test macro-F1 (0.3033). Despite achieving the highest accuracy (0.53), MNB predicts only 1 *maybe* correctly and classifies 58 of 68 *no* samples as *yes*, trading minority-class performance for majority-class accuracy. This illustrates precisely why macro-F1 is the appropriate metric: accuracy rewards majority-class prediction while hiding poor minority-class performance.

SMOTE oversampling (A2d) balanced training classes to 398 samples each. While CV macro-F1 on the resampled data was inflated (0.84), the test macro-F1 of 0.3544 shows synthetic oversampling does not translate to substantially better

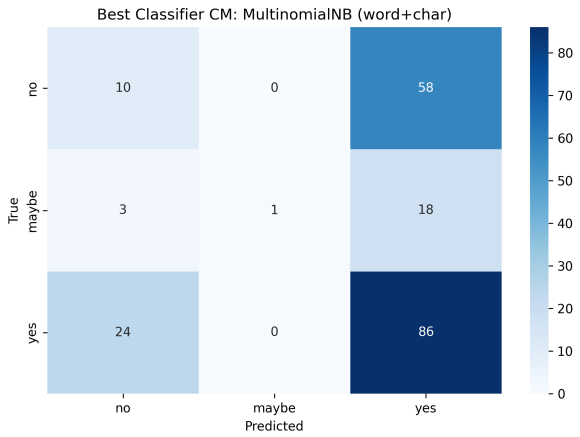

Figure 7: Best Axis 2 classifier confusion matrix (MNB). Despite 53% accuracy, only 1 *maybe* is correctly predicted, and 58 of 68 *no* samples are misclassified as *yes*.

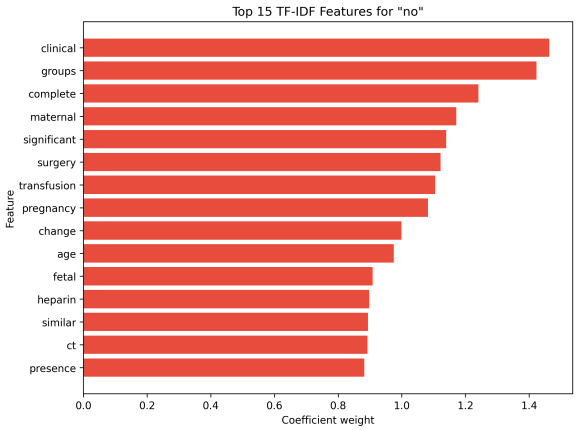

Figure 8: Top 15 TF-IDF features for the *no* class. Clinical trial terms (*clinical*, *groups*, *surgery*) dominate, reflecting topic-label correlations.

generalisation—the minority-class difficulty stems from genuine label ambiguity rather than insufficient training data.

The soft voting ensemble (A2e) achieved 0.2757—*worse* than all individual classifiers, because the models make highly correlated errors. The zero-shot BART-MNLI approach (A3) failed dramatically: test macro-F1 0.0633, accuracy 10.5%, predicting *maybe* for ∼95% of samples due to the NLI framing mismatch.

Fine-tuned BioBERT achieved the best overall test macro-F1 of **0.4108** (50% accuracy), benefiting from domain-specific pre-training and class-weighted loss (7; 6).

3.5 Top TF-IDF Feature Analysis

Figures 8, 9, and 10 display the top 15 discriminative TF-IDF features per class from the tuned LR model. The *no* class relies on clinical trial vocabulary: *clinical* (1.47), *groups* (1.42), *complete* (1.25). The *maybe* class is driven by topic-specific terms: *cycle* (1.42), *appendicitis*, *gastric cancer*—notably, these are medical topics rather than uncertainty indicators. The *yes* class features include result-reporting language: *cells* (1.18), statistical markers (*001*), and *showed*, *compared*. Critically, most features are topic-specific rather than decision-indicative, suggesting the model learns dataset-specific topic-label correlations rather than generalisable answer patterns.

3.6 Learning Curve Analysis

Figure 11 shows the learning curve for the tuned LR model. Training macro-F1 remains at 1.0 across all training set sizes, while validation macro-F1 starts at ∼0.28 and slowly increases to ∼0.37. This extreme gap reveals severe overfitting. The upward trend of the validation curve suggests more labelled data would continue to improve performance—the 1,000-sample expert-annotated PubMedQA subset is fundamentally small for learning robust decision boundaries across three classes.

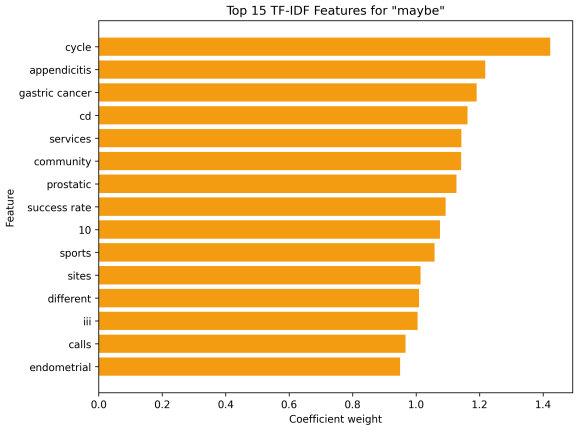

Figure 9: Top 15 TF-IDF features for the *maybe* class. Domain-specific terms (*cycle*, *appendicitis*, *gastric cancer*) drive predictions rather than uncertainty cues.

3.7 Overall Model Ranking

Figure 12 presents the complete ranking of all 12 model configurations. Fine-tuned BioBERT leads at 0.4108, followed by BioBERT embeddings+LR (0.3637) and SMOTE+LR (0.3544). The baseline and Axis 1 variants cluster around 0.33. The enriched Axis 2 classifiers score 0.28–0.30, and zero-shot BART-MNLI trails at 0.0633. Table 1 provides the full numerical summary.

Table 1: Summary of all experimental results, sorted by test macro-F1.

Model	CV F1	Test Acc	Test F1
BioBERT Fine-tuned	0.38	0.500	**0.4108**
BioBERT Emb+LR	0.37	0.465	0.3637
SMOTE+LR (w+c)	0.84	0.415	0.3544
Baseline: TF-IDF+LR	0.37	0.470	0.3306
Q+Ctx (Baseline)	0.37	0.470	0.3306
Q only	0.39	0.480	0.3302
Q+Ctx+Ans	0.38	0.460	0.3243
MNB (w+c)	0.36	0.530	0.3033
LR (w+c)	0.37	0.440	0.2831
Cal-SVM (w+c)	0.38	0.435	0.2806
Soft Voting Ensemble	0.29	0.420	0.2757
Zero-Shot BART-MNLI	—	0.105	0.0633

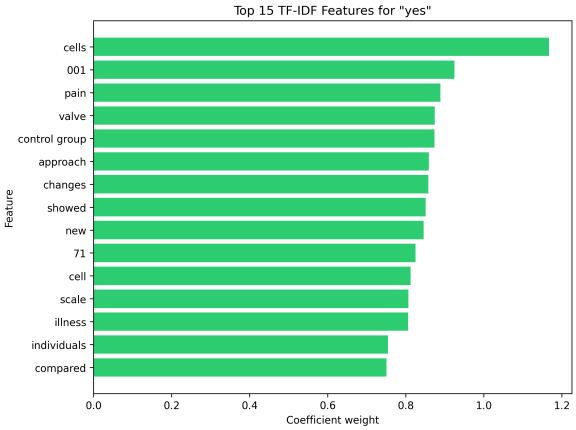

Figure 10: Top 15 TF-IDF features for the *yes* class. Result-reporting language (*cells*, *showed*, *compared*) and statistical markers (*001*) are prominent.

3.8 Error Analysis

3.8.1 Confusion Matrix and Per-class Error Rates

Figure 13 presents the confusion matrix for the best model (fine-tuned BioBERT). Compared to the baseline (Figure 3), BioBERT produces a more balanced distribution: 31 correct *no* (vs. 14), 4 correct *maybe* (vs. 2), and 65 correct *yes* (vs. 78). The trade-off is clear—BioBERT sacrifices some majority-class accuracy for better minority-class recall.

Figure 14 quantifies the per-class error rates: *maybe* at 81.8%, *no* at 54.4%, and *yes* at 40.9%. This ordering reflects both class size and intrinsic ambiguity—the *maybe* label represents genuinely uncertain evidence that even human annotators may disagree on.

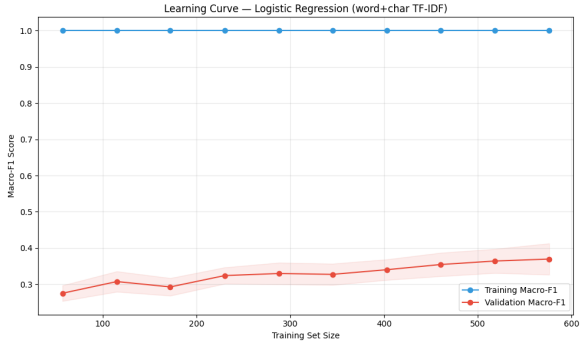

Figure 11: Learning curve for tuned LR (word+char TF-IDF). Training F1 stays at 1.0 while validation F1 slowly climbs to 0.37, indicating severe overfitting and potential for improvement with more data.

3.8.2 Misclassification Flows

Figure 15 shows misclassification flow directions. The largest flow is yes→no (39 samples), followed by no→yes (29) and maybe→yes (12). The bidirectional confusion between *yes* and *no* is particularly concerning for medical applications, as these represent opposite clinical conclusions. The maybe→yes flow (12 samples) reflects the model's tendency to resolve uncertainty towards the majority class.

3.8.3 Feature Distribution Analysis

Figure 16 compares question length distributions between correctly and incorrectly classified samples. The distributions overlap substantially, indicating text length is not a reliable error predictor. Figure 17 confirms the same finding for context length.

Figure 18 reveals a more nuanced picture: *strong*

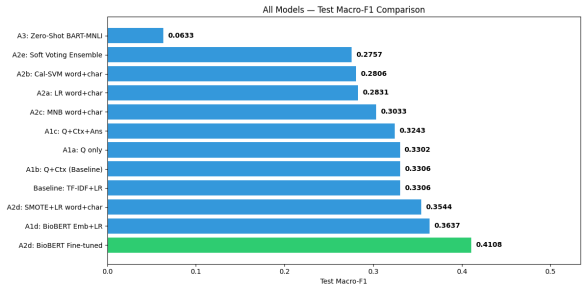

Figure 12: All models ranked by test macro-F1. Fine-tuned BioBERT (green) achieves the highest score of 0.4108. Zero-shot BART-MNLI trails far behind at 0.0633.

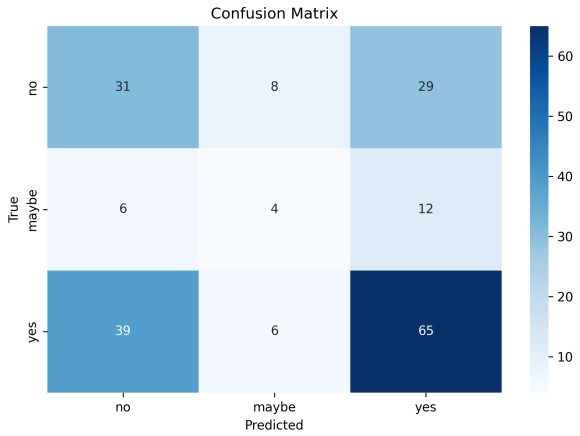

Figure 13: Best model confusion matrix (fine-tuned BioBERT). More balanced predictions than baseline: 31 correct *no*, 4 correct *maybe*, 65 correct *yes*.

evidence language is slightly more prevalent in correctly classified samples (0.20 vs. 0.15), while hedging appears at similar rates across both groups ($\sim$0.36). This suggests hedging language occurs across all three label classes and is therefore unreliable as a discriminator for *maybe*.

3.8.4 Error Overlap Between Models

Figure 19 presents the Jaccard similarity of error sets between the top four models. BioBERT fine-tuned and BioBERT embeddings share only 48% of errors, suggesting they capture different aspects despite using the same underlying architecture. SMOTE+LR and baseline share 74%, confirming SMOTE does not fundamentally change which samples are difficult. Critically, 24% of test samples (48 out of 200) are misclassified by *all* models, representing genuinely ambiguous cases.

3.8.5 Statistical Significance

McNemar's test between the baseline and fine-tuned BioBERT yields $p = 0.54$, indicating the

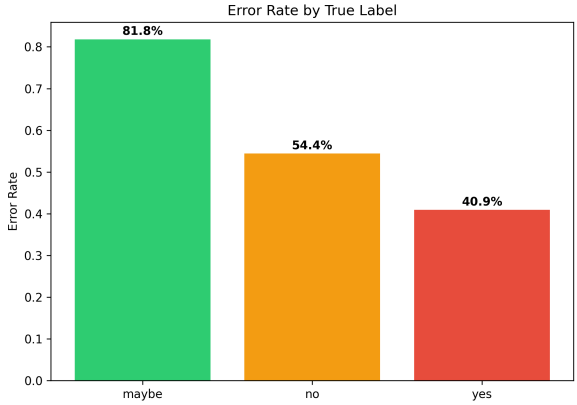

Figure 14: Error rate by true label. The *maybe* class has the highest error rate at 81.8%, followed by *no* at 54.4% and *yes* at 40.9%.

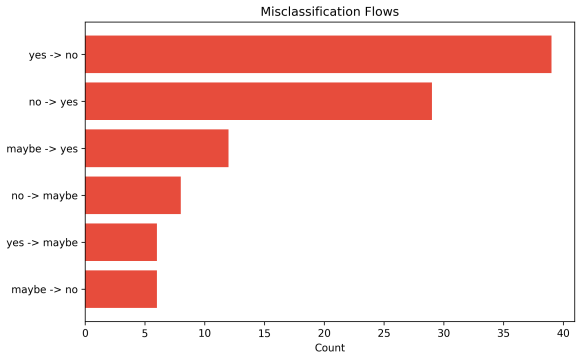

Figure 15: Misclassification flows. The largest error is yes→no (39 samples). Bidirectional yes/no confusion is the dominant error pattern.

improvement is **not statistically significant** at the 0.05 level. On a test set of 200 samples, the observed differences could arise from random variation.

3.9 Quality Predictor Results

Figure 20 shows the Random Forest feature importances. Context length dominates (0.32), followed by long answer length (0.28) and question length (0.19). Handcrafted linguistic features contribute minimally. Despite these informative features, all three quality predictor models achieved test macro-F1 of only 0.50–0.54—barely above random chance. Adding BioBERT embedding cosine similarity did not improve prediction. The correctness of predictions cannot be reliably forecast from surface-level text properties; the difficulty is semantic and domain-specific.

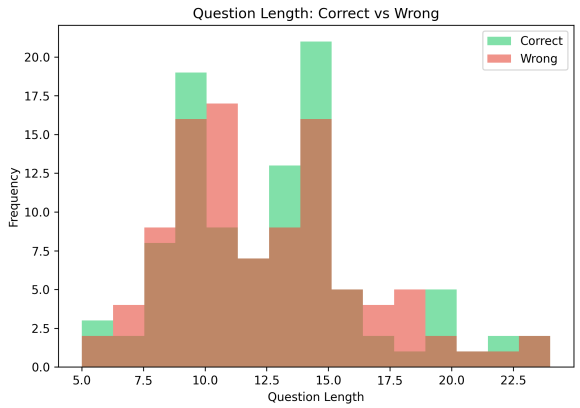

Figure 16: Question length: correct vs. wrong predictions. Distributions overlap heavily, confirming length is not a reliable error predictor.

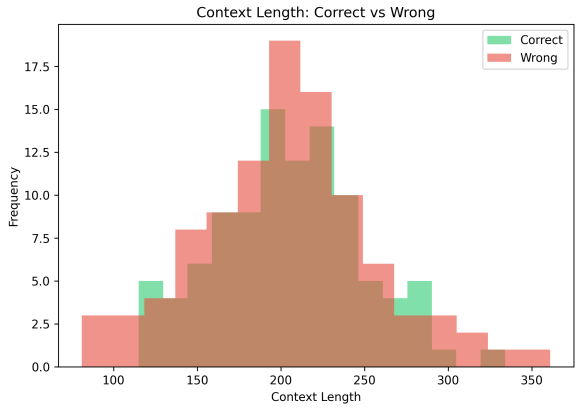

Figure 17: Context length: correct vs. wrong predictions. Again, substantial overlap indicates context length does not predict correctness.

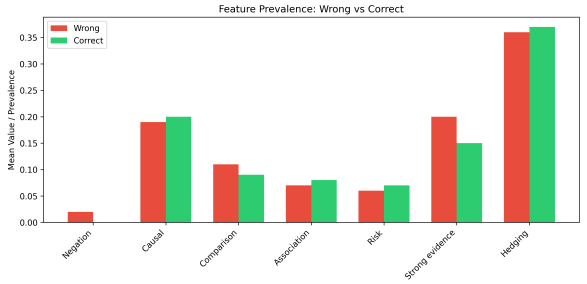

Figure 18: Handcrafted feature prevalence in wrong vs. correct predictions. Hedging appears equally in both groups; strong evidence is slightly more common in correct predictions.

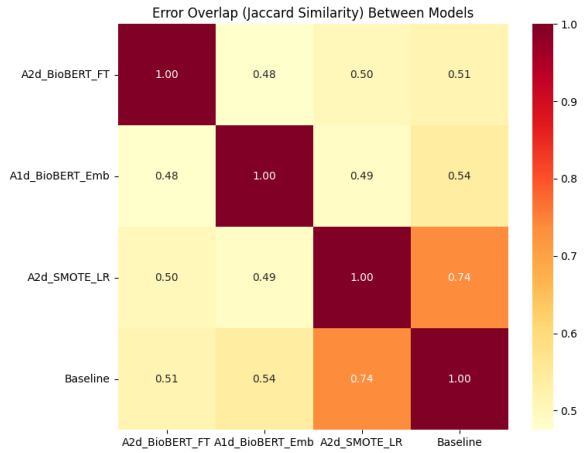

Figure 19: Error overlap (Jaccard similarity) between the top four models. SMOTE+LR and baseline share 74% of errors; BioBERT fine-tuned shares only 48–51% with other models.

4 Discussion

Our results reveal several important insights about biomedical QA classification.

The "maybe" class is fundamentally different. Unlike *yes* and *no*, which represent clear evidential conclusions, *maybe* captures epistemic uncertainty—insufficient evidence. This is not a "middle ground" but a qualitatively different judgement requiring meta-reasoning about evidence quality. The 81.8% error rate reflects this fundamental limitation of current models.

Topic-label correlations dominate over reasoning. The TF-IDF feature analysis reveals top features are medical topics (*clinical*, *gastric cancer*, *cells*) rather than reasoning indicators. The models exploit topic-label correlations in the training set rather than learning to evaluate evidence quality, which is a form of shortcut learning that limits generalisation.

The dataset size is a fundamental constraint. The learning curve shows no plateau in validation performance, and with only ∼79 *maybe* training examples, models have insufficient data to learn subtle patterns distinguishing uncertainty from certainty.

Supervised fine-tuning vastly outperforms zero-shot. The dramatic failure of BART-MNLI (0.0633 vs. 0.4108) demonstrates that general NLI capabilities do not transfer to domain-specific medical QA without adaptation. The NLI framing maps "uncertain medical evidence" to "neutral/maybe", ignoring the nuanced distinction between inconclusive and negative evidence (7).

Ensemble methods fail when errors correlate. The soft voting ensemble performed worse because all three classifiers operate on the same TF-IDF feature space and make similar mistakes. The Jaccard

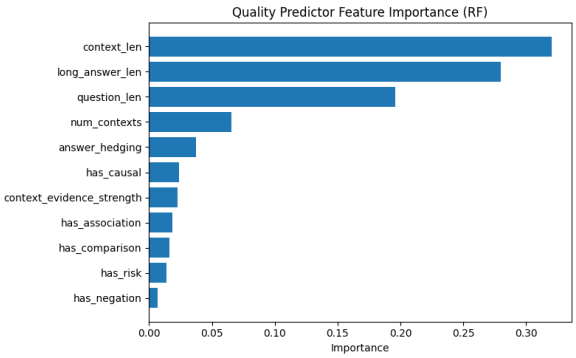

Figure 20: Quality predictor feature importance (Random Forest). Length-based features dominate, but overall predictive power remains near chance (∼0.52 macro-F1).

analysis (Figure 19) quantifies this: SMOTE+LR and baseline share 74% of errors. For ensembles to succeed, member models must be diverse in both architecture and feature representation—for example, combining a TF-IDF model with a BioBERT model might yield complementary error patterns.

Implications for clinical deployment. The bidirectional confusion between *yes* and *no* (Figure 15) is the most concerning error pattern from a clinical perspective. Misclassifying a *no* answer as *yes* could lead clinicians to believe a treatment is effective when evidence suggests otherwise, while the reverse could cause dismissal of effective interventions. In a deployed system, a confidence threshold that routes low-confidence predictions to human reviewers would be essential. However, our quality predictor results suggest that identifying such low-confidence predictions from input features alone is extremely difficult.

Comparison with prior work. Jin et al. (4) reported that fine-tuned BERT achieves approximately 55% accuracy on PubMedQA's expert-annotated test set. Our fine-tuned BioBERT achieves 50% accuracy (macro-F1 0.4108) on our 200-sample test split, which is broadly consistent given the different evaluation splits and our use of class-weighted loss that trades accuracy for balanced performance. The persistent difficulty of the *maybe* class aligns with their findings and suggests this is an intrinsic property of the task rather than a limitation of any particular approach.

Limitations

Several limitations should be noted. First, our test set contains only 200 samples (22 *maybe*), limiting statistical power—McNemar's test could not confirm significance even for the best model. Second, our BioBERT fine-tuning used a fixed hyperparameter configuration; systematic search over learning rates, batch sizes, and training epochs might yield further improvements. Third, we used a single random seed for data splitting, and results may vary across different splits. Fourth, our zero-shot experiment used only one LLM (BART-MNLI); larger models such as GPT-4 or domain-specific instruction-tuned models might perform better in the zero-shot setting, though this was beyond our computational budget.

5 Conclusions

This study systematically investigated text representation and classifier design for medical QA on PubMedQA. Our key findings:

Text representation has limited impact for traditional models. All TF-IDF variants achieved ∼0.33 test macro-F1. BioBERT embeddings offered a modest improvement (0.3637), indicating the bottleneck lies in the classification task.

Fine-tuned BioBERT is the strongest approach at 0.4108 test macro-F1, though McNemar's test ($p = 0.54$) shows this is not statistically significant on our 200-sample test set. All models struggle with the *maybe* class (81.8% error rate).

Zero-shot LLM classification fails (0.0633 macro-F1) due to the NLI framing mismatch. SMOTE and ensembles do not help—SMOTE shows inflated CV (0.84) but poor generalisation (0.3544), and the ensemble (0.2757) underperforms individuals due to correlated errors.

24% of errors are irreducible: 48 of 200 test samples are misclassified by every model, and the quality predictor confirms these cases involve inherent medical uncertainty that cannot be resolved from text alone.

Open challenges include ordinal regression to exploit the natural ordering of certainty, leveraging the 211k unlabelled PubMedQA instances via semi-supervised learning, retrieval-augmented generation (8) grounding classification in external evidence, and explainability methods to determine whether BioBERT genuinely evaluates evidence or exploits surface correlations. Multi-task learning that jointly predicts the decision label and generates the long answer could provide richer train-

ing signals. Finally, the recently available large language models with instruction tuning may perform better in zero-shot or few-shot settings than the BART-MNLI model we evaluated, and this direction warrants investigation with appropriate domain-specific prompting strategies.

References

[1] Qiao Jin, Zheng Yuan, Guangzhi Xiong, Qianlan Yu, Huaiyuan Ying, Chuanqi Tan, Mosha Chen, Songfang Huang, Xiaozhong Liu, and Sheng Yu. Biomedical question answering: A survey of approaches and challenges. *ACM Computing Surveys*, 55(2):1–36, 2022.

[2] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020.

[3] George Paliouras and Anastasia Krithara. BioASQ: A challenge on large-scale biomedical semantic indexing and question answering. BioASQ Workshop slides, Valencia, 27 September 2013.

[4] Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William W. Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577, 2019.

[5] Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. ClinicalBERT: Modeling clinical notes and predicting hospital readmission. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL) Workshop Track*, pages 1–9, 2020.

[6] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2021.

[7] Yifan Peng, Shankai Yan, and Zhiyong Lu. Transfer learning in biomedical natural language processing: An evaluation of BERT and ELMo on ten benchmarking datasets. In *Proceedings of the 18th BioNLP Workshop and Shared Task*, pages 58–65, 2019.

[8] Giacomo Frisoni, Miki Mizutani, Gianluca Moro, and Lorenzo Valgimigli. BIOREADER: a retrieval-enhanced text-to-text transformer for biomedical literature. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 5770–5793, 2022.

[9] Long N. Phan, James T. Anibal, Hieu Tran, Shaurya Chanana, Erol Bahıdıroğlu, Alec Peltekian, and Grégoire Altan-Bonnet. SciFive: a text-to-text transformer model for biomedical literature. *arXiv preprint arXiv:2106.03598*, 2021.

[10] Wonjin Yoon, Richard Jackson, Aron Lagerberg, and Jaewoo Kang. Sequence tagging for biomedical extractive question answering. *Bioinformatics*, 38(15):3794–3801, 2022.